Katarzyna Mroczkowska

Computer Vision Engineer

katarzyna.mroczkowska@example.com · Krakow, Poland

SUMMARY

Computer vision engineer working on defect detection for manufacturing lines. Eight years in applied machine learning, comfortable with the full loop from data labelling to edge deployment on small devices. I enjoy working with factory engineers and prefer pragmatic models that can run reliably in a noisy environment. I value reproducible training pipelines, honest evaluation and good communication between data, hardware and production teams.

SKILLS

PyTorch, OpenCV, YOLO, Segmentation, ONNX, TensorRT, Python, C++, Data labelling, Model quantisation, Albumentations, Docker

EXPERIENCE

Senior Computer Vision Engineer, Optiforge Manufacturing Tech 2019 – Present

- Trained a segmentation model detecting surface defects with 97% recall on production images, and the same approach was later reused by two other production lines without major changes.
- Exported models through ONNX and TensorRT to run at 30 frames per second on edge GPUs, which meant the models could run offline on the shop floor, away from any network.
- Designed the annotation guidelines and QA process for a team of six labellers, and agreed the acceptance criteria with quality engineers before training began.
- Reduced false rejects by 45% through targeted hard negative mining.
- Presented model performance to plant managers every month and adjusted thresholds based on their feedback.

Machine Learning Engineer, Wawel Imaging Systems 2016 – 2019

- Built classical and CNN based pipelines for medical image triage.
- Wrote C++ preprocessing modules.
- Developed a classification model that cut manual review of scans by 60%.
- Documented data collection protocols for radiologists.
- Optimised image preprocessing in C++, reducing latency per scan from 120 ms to 45 ms.

Research Assistant, Krakow Polytechnic Institute 2013 – 2016

- Worked on feature extraction for industrial inspection images.
- Wrote MATLAB and later Python prototypes.
- Co-authored two workshop papers.
- Evaluated open datasets for transfer learning and compared several backbone architectures.
- Presented results to clinicians and incorporated their feedback.

EDUCATION

- MSc Automation and Robotics, Krakow Polytechnic Institute (2014 – 2016)

LANGUAGES

Polish (native), English (C1)